

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

**Custom Cake Ordering Platform
with Smart Scheduling and Sales
Analytics**

Digital Dealers

Fernandez, Criss Angiela
Mandado, Keziah
Marcelo, Ziny Jen
Suamen, Maximo III
Sufeciencia, Joshua

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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CHAPTER 1

1.1 Background of the study

In today's fast-paced world, many businesses are shifting to online platforms to keep up with the growing expectations of their customers. The food service industry, particularly bakeries, is no exception. As more people turn to the internet for shopping and ordering, bakeries that still rely on manual and traditional processes are finding it harder to meet customer demands efficiently.

C and N Bakery is a local bakery that offers a wide range of products such as cakes, pastries, and custom cake orders. The bakery also accepts pre-orders for customers who need cakes for special occasions. Like many small businesses, it currently handles orders manually through walk-ins, phone calls, or personal messages without any digital system to support the process.

The process of ordering custom cakes is inherently more complex than standard retail transactions. It requires precise documentation of flavors, tiers, icing designs, and dedication messages, as well as strict adherence to pickup schedules to ensure product freshness. Relying on handwritten logs or scattered chat messages increases the risk of "order leakage"—where details are forgotten or schedules overlap—leading to wasted ingredients and disappointed customers.

Furthermore, the Philippine business landscape is rapidly evolving with the rise of e-wallets and digital banking. Customers now expect "frictionless" transactions where they can confirm an order and settle payments through QR codes or bank transfers instantly. For C and N Bakery, staying manual means missing out on a larger market of tech-savvy consumers who prioritize convenience and transparency.

This setup has caused several issues for both the customers and the staff. Customers find it inconvenient to place orders, especially for custom cakes, because there is no online catalog or ordering form they can access. On the other hand, the staff

struggles with keeping track of multiple orders and managing pickup schedules by hand, which often leads to confusion and mistakes.

These problems show the need for a better and more organized system. This study aims to develop a Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics, a web-based system that will make ordering easier for customers, help the staff manage schedules more efficiently, and give the bakery owner useful insights about their sales and business performance.

1.2 Statement of the problem

The bakery's current way of handling orders and schedules is done manually, which creates problems for both the customers and the staff. Without a proper digital system, the bakery faces recurring issues that affect its overall efficiency and the quality of service it provides.

The study seeks to address the following specific problems:

1. Customers have difficulty placing custom cake orders because the bakery does not have an online catalog or ordering platform where they can browse products, choose their cake preferences, and submit their orders conveniently.
2. The manual handling of pre-orders and pickup schedules is time-consuming and often leads to errors and miscommunication regarding specific cake details, making it hard for the staff to stay organized and for customers to secure their preferred order dates.
3. Customers have no way to check the status of their orders, and the bakery owner has no tool for monitoring sales data and business performance, making it difficult to make informed decisions.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics that will improve the bakery's ordering process, scheduling system, and business monitoring through the use of a web-based application.

1.3.2 Specific Objectives

Specifically, this study aims to:

- To develop an online cake catalog and custom cake ordering interface where customers can browse available products, customize their cake orders based on their preferences such as size, flavor, layers, frosting, toppings, and dedication messages, and get an automatic price calculation.
- To build a smart pre-order and scheduling system that includes an availability calendar, automated notifications via Email, and scheduling controls for staff to confirm and manage orders more efficiently.
- To create a real-time order tracking feature that lets customers see the current status of their order, along with an admin dashboard that shows sales reports and performance data to help the bakery owner monitor the business.
- To integrate multiple payment options including QR code payments, Bank Transfers, and Cash on Delivery (COD), allowing customers to choose the payment method that works best for them while providing the bakery with secure transaction records.

1.4 Scope and Delimitation

Scope:

This study focuses on the design and development of a web-based ordering platform for C and N Bakery. The system will allow customers to browse the bakery's products, place regular and custom cake orders, choose a pickup date through an availability calendar, pay online via QR code, Bank Transfer or Cash on Delivery, and track their order status in real time. Customers will also receive automated Email notifications for order confirmations and pickup reminders.

On the admin side, the system will allow the bakery owner and staff to manage incoming orders, confirm schedules, and view a sales analytics dashboard that includes reports and performance metrics.

The system is intended for two types of users: customers who will use the platform to order and track their cakes, and administrators who will oversee the operations and monitor sales through the system.

Delimitation:

The system will not include inventory management, payroll, supplier management, and human resource-related features. Access will be limited to web browsers only, with no mobile application available. Third-party delivery tracking will also not be supported, as the bakery operates on a personal pickup and in-shop delivery basis.

1.5 Significance of the Study

This study is significant to the following individuals and groups:

Customers. With the new system, customers will be able to place and customize their cake orders online, track the progress of their orders in real time, and receive notifications about their orders making the whole experience much more convenient and satisfying.

Bakery Owner (Administrator). The sales analytics dashboard will give the bakery owner a clearer picture of how the business is performing, which products are selling well, and how revenue is trending over time, helping them make better business decisions.

Bakery Staff. The smart scheduling system will take away a lot of the stress that comes with manually managing orders and schedules. Staff will be able to confirm and organize orders more easily, reducing the chance of mistakes.

The Organization. The bakery as a whole will benefit from a more organized and efficient operation. With fewer errors and happier customers, the business is better positioned to grow and compete in the local market.

Researchers. This study can serve as a reference for other researchers who are interested in developing similar digital solutions for small businesses, especially in the food service and bakery industry.

Future Developers. The system developed in this study can serve as a foundation or reference for future developers who want to build on or improve the platform for similar business needs.

1.6 Definition of Terms

The following terms are defined based on how they are used in this study:

Availability Calendar – A scheduling feature in the system that shows which dates are open or unavailable for pre-orders, so customers can pick a pickup date based on what the bakery can accommodate.

Custom Cake – A cake that the customer personalizes by choosing specific details like size, flavor, number of layers, frosting, design, and a dedication message.

Dashboard – The admin interface in the system that shows a summary of orders, sales, and other important business information in one place.

Online Catalog – The section of the platform where the bakery's products are displayed, including featured items.

Performance Metrics – Numbers and data tracked by the system that reflect how the business is doing, such as total orders, total sales, and which products are most popular.

Pre-Order – An order that a customer places ahead of time for a specific pickup date, which the staff then reviews and confirms based on availability.

QR Code Payment – A payment method where the customer scans a QR code using their phone to pay for their order digitally without needing cash.

Multi-Channel Payment – A system that offers various ways to pay, such as digital (QR), electronic (Bank Transfer), and manual (Cash on Delivery), to provide flexibility to the customer.

Bank Transfer – An electronic payment method where a customer sends money directly from their bank account to the bakery's account, with the transaction verified by the admin through the dashboard.

Real-Time Order Tracking – A feature that lets customers see exactly where their order is in the process, using a visual timeline with stages like Preparing, Baking, Packing, and Delivery.

Sales Analytics – The process of collecting and analyzing the bakery's sales data through the system to identify trends and generate useful business reports.

Smart Scheduling – An automated feature of the system that checks order availability, organizes baking and pickup schedules, and sends reminders to both customers and staff.

Email Notification – A system-generated message automatically sent to customers to provide updates on order status, including order confirmation, down payment verification, and pickup or delivery readiness.

Web-Based System – A system that runs on a web browser and can be accessed online without needing to install any software on the user's device.

CHAPTER 2

2.1 Foreign Studies

Lim and Hamzah (2023) confirm that transitioning from fragmented manual methods—such as instant messaging and Microsoft Excel—to a dedicated web-based platform significantly improves operational transparency. Their research emphasizes that a centralized database allows for the systematic storage of order details, which prevents the data loss often associated with manual record-keeping. The system was designed with distinct modules for customers to browse categories and track order statuses, while the admin module featured a dashboard for managing product availability and generating sales reports. The study's User Acceptance Testing (UAT) phase involved rigorous evaluation by both business owners and end-users, focusing on usability and functionality. The findings showed that the system effectively addressed "bottlenecks" in the order fulfillment process, particularly during peak seasons when manual systems typically fail to handle high volumes. By implementing automated order confirmations and a clear digital queue, the researchers concluded that the system not only improved "business smoothness" but also increased customer trust through professionalized digital interactions. This study is relevant because it explains why switching from manual phone calls and walk-ins to a web platform will make C and N Bakery's operations much clearer and more organized.

Rajaput et al. (2026) addresses the operational inefficiencies inherent in traditional, manual cake ordering processes. The researchers identify that reliance on handwritten registers and verbal communication frequently leads to critical errors, such as incorrect cake weights, misspelled names for icing, and scheduling conflicts regarding delivery dates. To mitigate these risks, the study proposes a web-based digital customization module that categorizes orders into specific data fields, including flavor, occasion type, and personalized

messaging. The study demonstrates that by utilizing a Waterfall Methodology to build a structured database, bakeries can ensure that every customer requirement is captured accurately and stored for real-time administrative review. Furthermore, the findings highlight that the transition from manual logs to a centralized digital framework significantly reduces human error and improves the speed of order processing. The research concludes that an intuitive, computerized management system is a vital tool for small-scale bakeries to maintain data integrity and enhance customer satisfaction in an increasingly digital marketplace. This study is relevant because it justifies why C and N Bakery needs a specific customization menu to help avoid common mistakes like wrong cake flavors or misspelled names on the icing.

Hsu et al. (2024) investigated consumer perceptions toward purchasing bakery goods online using an extended Technology Acceptance Model (TAM). Unlike general food studies, this research specifically focused on the "sensory gap" of buying bread and cakes online, where customers cannot smell or touch the product. The findings revealed that Computer Self-Efficacy (how confident a user feels using the app) and Subjective Norms (social influence) significantly impact how useful and easy the system is perceived to be. The study emphasizes that for traditional bakery owners to successfully move online, they must provide "high-quality visual information" and a seamless interface to make up for the lack of physical interaction. This research is relevant because it highlights the importance of including high-resolution photos and clear descriptions in the C and N Bakery catalog to give customers the confidence to order online.

Prawirayudha et al. (2025) investigated the impact of QR code technology on customer satisfaction and repurchase intentions within the food and beverage industry. The study specifically analyzed three key dimensions: Perceived Usefulness, Perceived Ease of Use, and Perceived Flow. The researchers established that when customers perceive digital systems as easy to navigate, it creates a "flow" state that

enhances overall satisfaction and encourages them to return. The study concludes that the digitalization of payments is a critical innovation for customer loyalty. This research is relevant because it validates the goal of integrating modern payment options—including QR codes and bank transfers—into the C and N Bakery platform. It proves that by making transactions faster and more convenient.

Azhari and Sutarman (2024) demonstrate how digital integration modernizes small businesses through a real-time system. To fix the problems of fragmented data, the researchers utilized Firebase as a Backend-as-a-Service (BaaS) to ensure that transactions are synced instantly across the platform. This architecture provides high-speed data integrity—which is essential for businesses like bakeries—without the complexity of traditional, manual server management. The study concludes that moving to an automated, real-time data flow significantly improves operational performance and helps owners move away from manual record-keeping. This research is relevant because it confirms that using Firebase is a smart technical choice for C and N Bakery, as it allows order updates and sales data to be synced instantly without a complicated setup.

Ariani, Audy, and Putra (2025) examines the impact of Firebase Authentication on data confidentiality and system integrity. The researchers highlight that Firebase's authentication service provides a critical security layer by facilitating efficient login, registration, and token verification systems. These mechanisms protect the confidentiality of user data, such as personal history and transaction records, by ensuring that only authorized individuals can access specific datasets. The study further notes that Firebase's performance is optimized for real-time environments, with average response times of 120 ms for data reads and 140 ms for data writes, ensuring that security checks do not hinder the user experience. By utilizing token-based validation, the researchers successfully mitigated unauthorized access attempts, confirming that

Firestore offers a fast and secure infrastructure suitable for businesses requiring reliable data management. The conclusion emphasizes that for small-scale operations, this integrated security model significantly reduces the risk of data breaches while maintaining operational efficiency. This study is relevant because it explains why Firestore Authentication is the right fit for C and N Bakery, proving that it provides strong security to keep customer data safe without making the website feel slow or laggy.

Pargaonkar (2023) evaluates the operational suitability of SDLC models within software quality engineering. The study establishes that while the Waterfall model remains effective for projects with fixed, well-defined requirements, the Agile methodology is the superior choice for dynamic, user-centered applications where requirements are expected to evolve during the development process. The research emphasizes that Agile's iterative nature facilitates continuous feedback and incremental delivery, which significantly reduces the risks associated with late-stage requirement changes. Pargaonkar concludes that for projects requiring high levels of stakeholder interaction—such as a localized ordering platform—the Agile framework provides the necessary flexibility to adapt to user feedback without compromising the overall project quality or timeline. This study is relevant because it justifies why we focused on constant communication with the owner during development. Instead of guessing what would work, the Agile method allowed for a back-and-forth conversation to ensure the features actually solve the bakery's daily struggles.

2.2 Local Studies

Bermudez et al. (2022) examined the operational inefficiencies of a local bakery in the Philippines caused by manual data handling. The research highlighted that the absence of an automated system led to slow transaction speeds, specifically noting that the manual process of writing orders and calculating totals caused significant delays during peak hours. The authors observed that manual record-keeping made it difficult for the owner to generate accurate daily sales summaries, often resulting in "data discrepancies" between the physical logbooks and actual cash on hand. To address this, the study developed a centralized platform to ensure that every transaction was recorded instantly and accurately. The findings emphasized that for small-scale bakeshops, digital transformation is essential for organizing order data, improving the speed of service, and ensuring long-term business sustainability. Similar to the foreign study of Lim and Hamzah (2023), this local research confirms that manual record-keeping is a major problem for Filipino bakeries. This is relevant to the current project because it justifies the need for C and N Bakery to have a web-based system that organizes order data and provides clear sales reports without the errors of manual calculation.

Li (2024) addresses the operational challenges faced by Philippine eateries still utilizing manual recording methods. The research highlights that traditional systems involving pens and calculators for order tabulation are insufficient for today's fast-paced environment. To resolve this, the researcher developed a digital ordering layout featuring an efficient interface designed to enhance user engagement for both delivery and self-collection services. The findings suggest that transitioning to a structured digital management style allows proprietors to manage menus and customer orders more effectively. Similar to the foreign findings of Rajput et al. (2026), this local study confirms that relying on manual notes is no longer enough for Filipino businesses. This is relevant to the project because it supports the goal of replacing C and N

Bakery's manual logs with an online platform that makes organizing custom orders much easier for the staff.

Lambino and Murcia (2025) utilized path analysis to examine the factors influencing the adoption of online food ordering systems among 367 consumers in Davao City, Philippines. The research focuses on why Filipino customers choose to use platforms like GrabFood and FoodPanda, evaluating variables such as System Quality, Perceived Usefulness, and User Convenience. The findings reveal that System Quality and Perceived Usefulness are the most significant determinants of a positive user attitude and long-term loyalty. The study highlights that a high-functioning system—one that provides clear information and ease of navigation—is essential for fostering trust in the competitive Philippine food delivery market. The researchers conclude that for local businesses to sustain customer engagement, they must prioritize technological improvements that enhance the perceived ease-of-use, as functionality plays a critical role in whether a customer continues to utilize a digital platform or returns to traditional ordering methods. Similar to the foreign study of Hsu et al. (2024), this local research explains why the C and N Bakery platform must be easy to use as these factors directly determine whether a Filipino customer will return to the site or go back to traditional ordering.

Espinosa, Quiambao, and Ramos (2023) investigated why 285 Filipino small business owners are moving toward digital payments. Using the UTAUT and DOI theories, they found that "Relative Advantage" is the biggest reason to switch, as it reduces the risks of handling physical cash and avoids the need for expensive equipment. The study highlighted that simple tools, like static QR codes, appeal to entrepreneurs looking for low-complexity ways to manage money. The researchers concluded that digital payment integration is a strategic necessity to align with the habits of modern Filipino consumers. Similar to the foreign study of

Prawirayudha et al. (2025), this local research justifies the need for C and N Bakery to offer various payment options—including QR codes and bank transfers. It proves that providing these digital choices is a smart move to enhance security, reduce cash-handling errors, and meet the digital preferences of local buyers.

Manaligod et al. (2022) developed a web-based management system for local service providers in the Philippines to address the issues of slow, manual data processing. Their research focused on creating a centralized platform where transaction data is updated in real-time to ensure the owner has the most current information. The study emphasized that a web-based approach allows for secure user authentication and easier data retrieval compared to traditional paper logs. The findings showed that by moving to a digital system, the business could handle more orders with fewer errors. Similar to the foreign findings of Azhari and Sutarman (2024), this local research explains why a real-time digital system is a huge upgrade for small businesses. This is relevant to the project because it supports the use of Firebase Realtime Database and Authentication, ensuring that C and N Bakery's orders are synced instantly and their records remain secure.

Tii Pen (2023) developed a web-based management system for a local retail store in the Philippines to modernize their manual operations. The study focused on creating a secure administrative dashboard that ensures only authorized users can access business sales data and inventory records. The findings highlighted that moving to a digital platform reduces the risk of data loss and improves the accuracy of transaction records compared to paper-based logs. Similar to the foreign study of Ariani et al. (2025), this local research explains why secure user access is necessary for a business. This is relevant to the project because it supports the implementation of Firebase Authentication to keep C and N Bakery's data private and secure from unauthorized access.

Aquino (2022) looked into how Filipino businesses are using digital tools to better understand what their customers actually want. The research found that using real-time systems allows shops to change and adapt quickly to keep up with customer expectations. The study highlights that a step-by-step digital shift is the best way for small businesses to keep their relationships strong while making sure the new tech actually fits their daily routine. Just like Pargaonkar (2023), this study shows why being flexible is important for small shops. It is relevant because it explains why we built the bakery's site piece-by-piece—allowing me to fix and improve the cake menu and QR payments until they were easy for everyone to use.

2.3 Related Literature

The architecture and operational framework of the Custom Cake Ordering Platform are grounded in several key areas of software engineering and digital business management. To ensure the system is both functional and adaptable, the project follows a descriptive-developmental research design. Pargaonkar (2023), in *A Review of Software Development Life Cycle (SDLC) Models: Comparison of Waterfall and Agile Methodologies* (International Journal of Science and Research, ISSN 2319-7064), establishes that while traditional models like Waterfall are effective for projects with fixed requirements, the Agile methodology is the superior choice for dynamic, user-centered applications. By adopting Agile, the researchers can build the system in iterative "sprints," allowing the bakery owner to test features like the cake customization menu early on. This iterative process ensures that the final product isn't just a generic website, but a tool that has been refined through constant feedback to solve the bakery's specific daily struggles.

In terms of technical infrastructure, the transition from manual logs to a digital environment requires a stable yet scalable backend. Azhari and Sutarman (2024), in *Modernizing Small Business Operations through Real-time Backend-as-a-Service (BaaS) Integration* (Journal of Computing and Software

Technology, ISSN 2722-2543), highlight that BaaS solutions provide high-speed data integrity without the heavy burden of manual server management. By integrating Firebase (Google Cloud, 2024) as the primary backend, the platform leverages cloud-based storage and hosting. This is a strategic move for small-scale enterprises like C and N Bakery because it offers a "serverless" environment that reduces overhead costs while maintaining professional-grade security and speed, ensuring that the website remains responsive even as the bakery's customer base grows.

The efficiency of handling complex orders, such as custom cakes with varying tiers and flavors, depends heavily on the underlying database structure. Labastida et al. (2024), in *Web-based Ordering System for Start-UP Business with Forecasting* (World Journal of Advanced Research and Reviews, ISSN 2581-9615), emphasizes that centralized digital platforms are essential for real-time data synchronization and accurate business forecasting. Unlike traditional paper-based logs which are prone to misinterpretation, a real-time NoSQL database allows for a flexible data schema. This means that as soon as a customer selects a specific frosting or dedication message, the data is pushed instantly to the admin dashboard. This eliminates the need for manual refreshes and allows the bakery staff to update order statuses—from "Baking" to "Ready for Pickup"—in a way that is immediately visible to the customer, thereby closing the communication gap.

Finally, to ensure the system is inclusive of the diverse payment habits in the Philippines, the platform incorporates a multi-channel payment framework. Espinosa, Quiambao, and Ramos (2023), in *A Study on the Adoption of Quick Response (QR) Code Payments Among Micro and Small Enterprises in the City of Manila* (Animo Repository, DLSU), establish that "Relative Advantage" and "Perceived Ease of Use" are the primary drivers for digital tool adoption among local entrepreneurs. By integrating QR codes, Bank Transfers, and Cash on Delivery (COD), the system caters to both tech-savvy users and traditional customers. This financial flexibility is further supported by Automated Email Notifications (Firebase Functions, 2024), which act as a digital bridge between the bakery and the client. These automated triggers provide instant confirmation and payment verification, which not only lightens the administrative load on the staff but

also builds significant consumer trust by providing transparent, real-time updates on every transaction.

2.4 Synthesis of the Reviewed Literature and Studies

The reviewed literature and studies provide a comprehensive foundation for the development of the Custom Cake Ordering Platform for C and N Bakery. Both foreign and local studies, such as those by Lim and Hamzah (2023) and Bermudez et al. (2022), emphasize that manual record-keeping in the food industry leads to data discrepancies and operational inefficiencies. This common theme justifies the necessity of moving C and N Bakery from a manual, chat-based ordering process to a centralized web-based system to ensure data integrity and better organization.

Furthermore, the research highlights the shifting behavior of modern consumers. Studies by Hsu et al. (2024) and Lambino and Murcia (2025) point out that high-quality visuals and system ease-of-use are the primary drivers of customer trust and loyalty. This is directly addressed in the proposed system through the development of a structured Online Catalog and a user-friendly Cake Customization interface. To support these features, the technical literature identifies the Agile Model and Firebase Cloud Architecture as the most effective tools for creating a responsive, real-time environment that can adapt to user feedback during development.

Finally, the inclusion of multi-channel payments—specifically QR codes, bank transfers, and Cash on Delivery (COD)—is backed by the findings of Espinosa et al. (2023), who argue that payment flexibility is a strategic necessity for local MSMEs in the Philippines. While existing systems often focus on simple product listings, the proposed platform fills a significant gap by combining customization, smart scheduling, and real-time status tracking into a single solution. Collectively, these studies and literature validate that the transition to a digital platform is not only a technological upgrade but a vital business strategy for C and N Bakery to improve its service quality and sustain growth in a competitive market.

CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

3.1 Research Design

Our study employed a **descriptive-developmental research design**.

Descriptive research was utilized to clearly identify and analyze the existing manual processes of C and N Bakery in handling custom cake orders, pre-orders, pickup scheduling, and sales monitoring. This involved gathering detailed information about the current problems through interviews with the bakery owner and staff, direct observation of daily operations, and a thorough review of related local and foreign studies on bakery and food ordering systems.

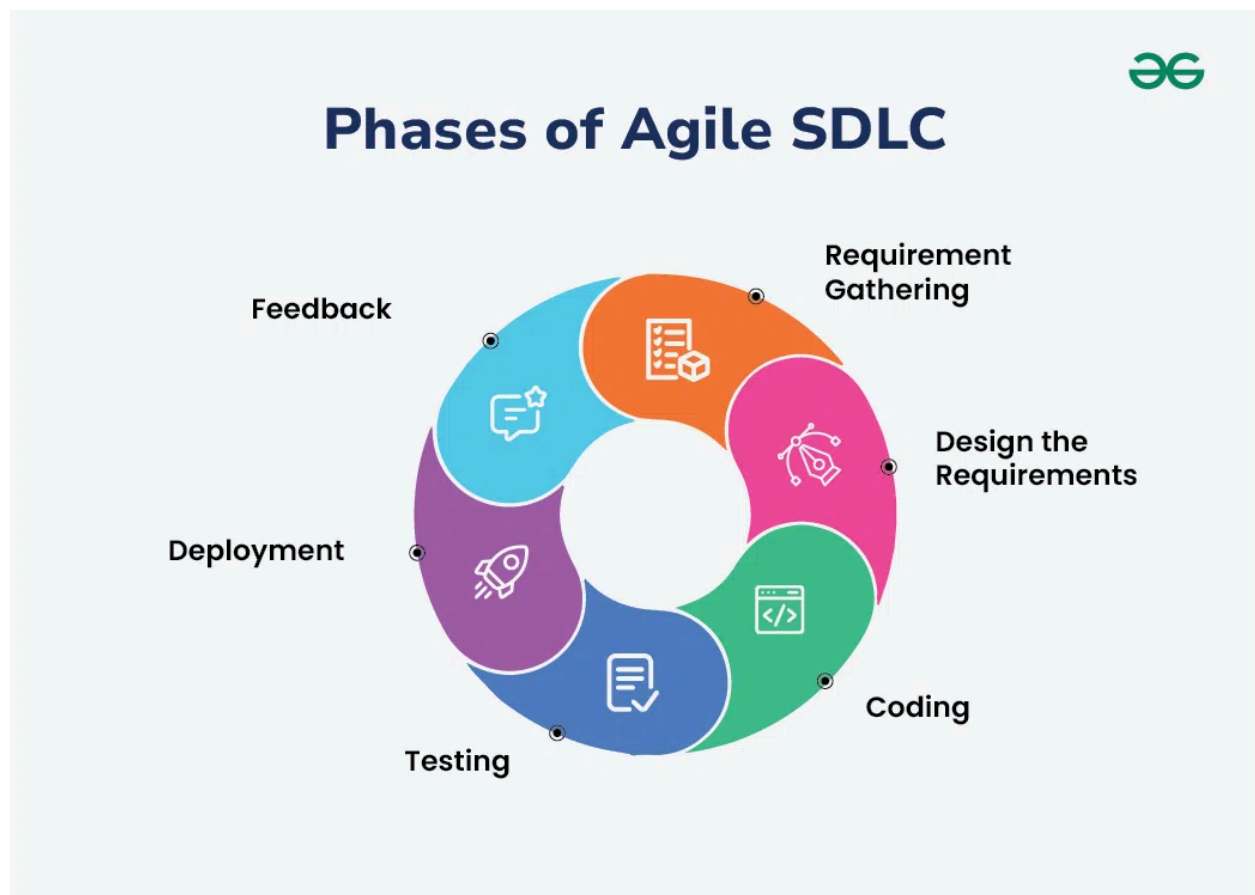
Developmental research formed the core of the study, focusing on the systematic design, development, and refinement of a functional web-based Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics. This approach is highly appropriate for technology-oriented capstone projects as it emphasizes the creation of a practical software solution to solve identified real-world problems. The developmental process followed a structured software development lifecycle to ensure the system is reliable, user-friendly, and tailored to the specific needs of the bakery.

Experimental research was not applicable in this study because the primary goal was not to establish cause-and-effect

relationships through controlled experiments or hypothesis testing, but rather to describe the existing situation and develop an integrated system solution.

3.2 System Development Methodology

The researchers adopted the **Agile Model** as the Software Development Life Cycle (SDLC) for this project. The Agile Model was chosen because it is iterative and incremental, allowing for flexibility, continuous feedback from the bakery owner and staff, and rapid adjustments during development. This is ideal for a small-scale web-based system where user requirements may evolve as the bakery tests early versions of the platform.



The development process followed these main phases:

- **Planning:** Identified project goals, scope, deliverables, and resources. Initial requirements were gathered from the bakery through interviews and observation.
- **Requirements Analysis:** Analyzed the functional and non-functional requirements, including custom cake ordering, smart scheduling, real-time tracking, email notifications, and sales analytics.
- **Design:** Created system architecture, database design, user interface wireframes, and detailed feature specifications.
- **Implementation (Development):** Coded the actual system in iterative sprints, developing features such as the customer ordering interface, admin dashboard, and scheduling calendar.
- **Testing:** Conducted unit, integration, system, and user acceptance testing in each iteration to identify and fix issues early.
- **Deployment:** Installed and configured the system at C and N Bakery for actual use.
- **Maintenance:** Provided post-deployment support, bug fixes, and possible enhancements based on user feedback.

3.3 System Architecture

The proposed system utilizes a **Serverless Cloud Architecture** powered by **Firebase**. Unlike traditional client-server models that require a manual back-end (like PHP), this architecture allows the front-end to communicate directly with cloud services.

- **Client Side:** A responsive web interface built with HTML5, CSS3, and Vanilla JavaScript.
- **Backend-as-a-Service (BaaS):** Firebase manages the logic, including:
 - **Authentication:** Handles secure customer and admin logins.
 - **Realtime Database:** Stores and syncs order data instantly.

3.4 Tools and Technologies Used

The following tools and technologies were utilized in developing the system:

- **Front-end Core: HTML & CSS:** For structuring and styling the user interface.
 - **Vanilla JavaScript:** For client-side logic, DOM manipulation, and integrating Firebase services without the overhead of external frameworks.
- **Platform-as-a-Service: Firebase:** Serves as the primary back-end infrastructure, providing a NoSQL database, user authentication, and hosting.
- **Database: Realtime Database:** A flexible, scalable NoSQL cloud database used to store customer profiles, custom cake orders, and sales analytics in real-time.
- **Development Tools: Visual Studio Code:** The primary Integrated Development Environment (IDE).
 - **Git/GitHub:** For version control and collaborative coding.
 - **Firebase Console:** For monitoring database traffic, managing users, and analyzing system logs.

3.5 Data Gathering Techniques

The researchers employed the following data gathering techniques to collect accurate and relevant information:

- **Interviews:** Conducted structured and semi-structured interviews with the bakery owner and staff to understand current manual processes, pain points, and desired features.
- **Observation:** Directly observed the bakery's daily operations, including order taking, scheduling, and sales recording, to identify inefficiencies.
- **Document Analysis:** Reviewed existing manual documents such as order forms, sales records, and scheduling notes.

These techniques ensured that the developed system directly addresses the real needs of both the bakery and its customers.

3.6 Testing Procedures

The system underwent rigorous testing to ensure quality and reliability:

- **Unit Testing:** Individual modules (e.g., custom cake price calculator, availability calendar, order submission) were tested separately to verify correct functionality.
- **Integration Testing:** Combined modules were tested to ensure they work together seamlessly (e.g., ordering interface connected to the scheduling system).
- **System Testing:** The complete integrated system was tested in an environment similar to actual use, covering all features end-to-end.
- **User Acceptance Testing (UAT):** The bakery owner, staff, and selected customers tested the system and provided feedback on usability and functionality. Adjustments were made based on their recommendations before final deployment.

3.7 Evaluation Criteria

The developed system was evaluated based on the following criteria:

- **Functionality:** The system performs all intended features correctly, including an online catalog (menu) for browsing available products, custom ordering with automatic pricing, smart scheduling, real-time tracking, automated email notifications for order confirmation, down payment verification, and pickup or delivery updates, and sales analytics.
- **Usability:** The interface is intuitive, easy to navigate, and user-friendly for both customers and administrators.
- **Reliability:** The system operates consistently without frequent crashes or data loss and handles errors gracefully.
- **Efficiency:** The system processes orders, calculations, and reports quickly, reducing manual workload and response time.
- **Security:** User data, orders, and payments are protected through proper authentication, input validation, and secure

practices (e.g., password hashing and secure QR code handling).

This chapter outlines the systematic approach used to design, develop, test, and evaluate the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics, ensuring the final product is practical and effective for C and N Bakery.

Appendix foreign studies

Lim, Y. Y., & Hamzah, M. H. I. (2023). The Development of Caketisserie Bakery & Café Online Cake Ordering System. *Applied Information Technology and Computer Science*, 4(2), 1412-1430. ISSN: 2716-1218

<https://publisher.uthm.edu.my/periodicals/index.php/aitcs/article/view/12088>

Karansing B. Rajaput, Priyanka S. Dhodamani, Sanika S. Suryawanshi, Sonali B. Sopnur, Shubhangi N. Kodag, & Komal K. Bamane. (2026). Bakery Shop Cake Customization Management System. *International Research Journal of Innovations in Engineering and Technology - IRJIET*, 10(2), 92-94. Article DOI

https://irjiet.com/common_src/article_file/IRJIET1020151771939174.pdf

Hsu, C. Y., Wang, M. Y., Chen, Y. Z., Chien, C. Y., & Lee, T. I. (2024). Exploring technology acceptance model to investigate consumers' online purchasing behavior of bakery: The case of the Taiwan. *Journal of Infrastructure, Policy and Development*, 8(9), 7414.

https://www.researchgate.net/publication/383901950_Exploring_technology_acceptance_model_to_investigate_consumers'_online_purchasing_behavior_of_bakery_The_case_of_the_Taiwan

Prawirayudha, Angbawangsa & Syahputera, Muhammad & Adisaputra, Raihan & Rahim, Rano. (2025). The Impact of QR Codes on Customer Satisfaction For Repurchase Intention on the Basis of Perceived Usefulness, Perceived Ease of Use and Perceived Flow in Food and Beverage Restaurant. *Jurnal Impresi Indonesia*. 4. 3823-3835. 10.58344/jii.v4i10.7026.

<https://www.researchgate.net/publication/396280519>

Azhari, F. N., & Sutarman. (2024). Transforming SME operations with real-time mobile POS and Firebase integration. *International Journal of Science, Technology, Engineering and Mathematics*, 4(4), 108-135.

<https://doi.org/10.53378/ijstem.353134>

Ariani, Audy & Putra, Agung & Sugata, Tri. (2025). IMPLEMENTATION OF FIREBASE IN THE DEVELOPMENT OF ANDROID-BASED QUEUE RESERVATION AND TREATMENT RECORD APPLICATIONS. *JSil (Jurnal Sistem Informasi)*. 12. 63-71. 10.30656/jsii.v12i1.9880.

https://www.researchgate.net/publication/390094294_IMPLEMENTATION_OF_FIREBASE_IN_THE_DEVELOPMENT_OF_ANDROID-BASED_QUEUE_RESERVATION_AND_TREATMENT_RECORD_APPLICATIONS

Pargaonkar, S. (2023). A comprehensive research analysis of software development life cycle (SDLC) agile & waterfall model advantages, disadvantages, and application suitability in software quality engineering. *International Journal of Scientific and Research Publications*, 13(8), 110-116. <http://dx.doi.org/10.29322/IJSRP.13.08.2023.p14015>

Appendix Local studies

Bermudez, Dorren, et al. "Information System Development Plan for the Berna's Bakeshop." *International Journal of Innovations in Engineering Research and Technology*, vol. 8, no. 06, 2021, pp. 441-453.
<https://www.neliti.com/publications/429262/information-system-development-plan-for-the-berna-s-bakeshop#cite>

Li, J. (2024). Development of a web-based food ordering system. *Asia Pacific Journal of Management and Sustainable Development*, 12(2), Part 3.
<https://www.ejournals.ph/article.php?id=25951>

Lambino, J., & Murcia, J. V. (2025). A path analysis of factors that influence acceptance of online food ordering systems among Davao City customers. *Business and Organization Studies E-Journal*, 3(1), 1-15.
<https://ieesjournals.com/index.php/bosej/article/view/221>

Espinosa, J. S., Quiambao, M. B., & Ramos, V. V. (2023). A study on the adoption of quick response (QR) code payments among micro and small enterprises in the City of Manila as mediated by the behavioral intention to adopt and moderated by age and income level.
https://animorepository.dlsu.edu.ph/etdb_acc/44/

Tii Pen. (2023). *Web-based management system with POS for Cultivate*. Scribd.
<https://www.scribd.com/document/685013898/Web-Based-Management-System-With-POS-for-Cultivate>

Aquino, R. (2022). Customer relationship management systems and the mediating role of digitalization in Philippine marketing. *International Journal of Social Science and Humanities Research (IJSSHR)*, 7(12).
<https://ijsshr.in/v7i12/Doc/71.pdf>